A Meta-Analysis on the Relation Between Executive Functioning and Second Language Reading

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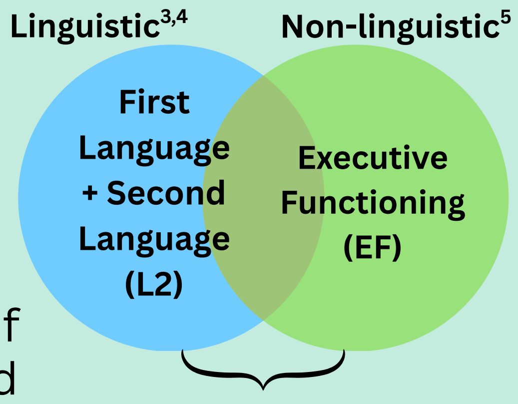
Introduction

Reading Comprehension (RC) requires joint skills:



- Reading is an active process of constructing and revising mental representations, requiring the coordination of dynamic systems[1,2,3]

- EF skills particularly important for bilingual readers due to high demands on co-active linguistic systems[6,7,8,9]
- This meta-analysis investigates potential associations between three core EFs and L2 RC across existing works

Method

Literature Search

6,615 identified through databases
→ **4,793** after duplicates removed
→ **694** after abstract screening
→ **54** after full text screening
→ **28** included in meta-analysis

28 studies from Jan. 1900 to May 2025 (**32** independent samples, N = 4,274, M_{age} = 13.9) met the following criteria:

- ☑ Is an empirical study
- ☑ Contains at least one RC measure
- ☑ Contains at least one EF measure
- ☑ Contains bilingual samples
- ☑ Contains neurotypical samples
- ☑ Reports raw data, correlation matrices, or covariance matrices

Variables & Measures

L2 RC:

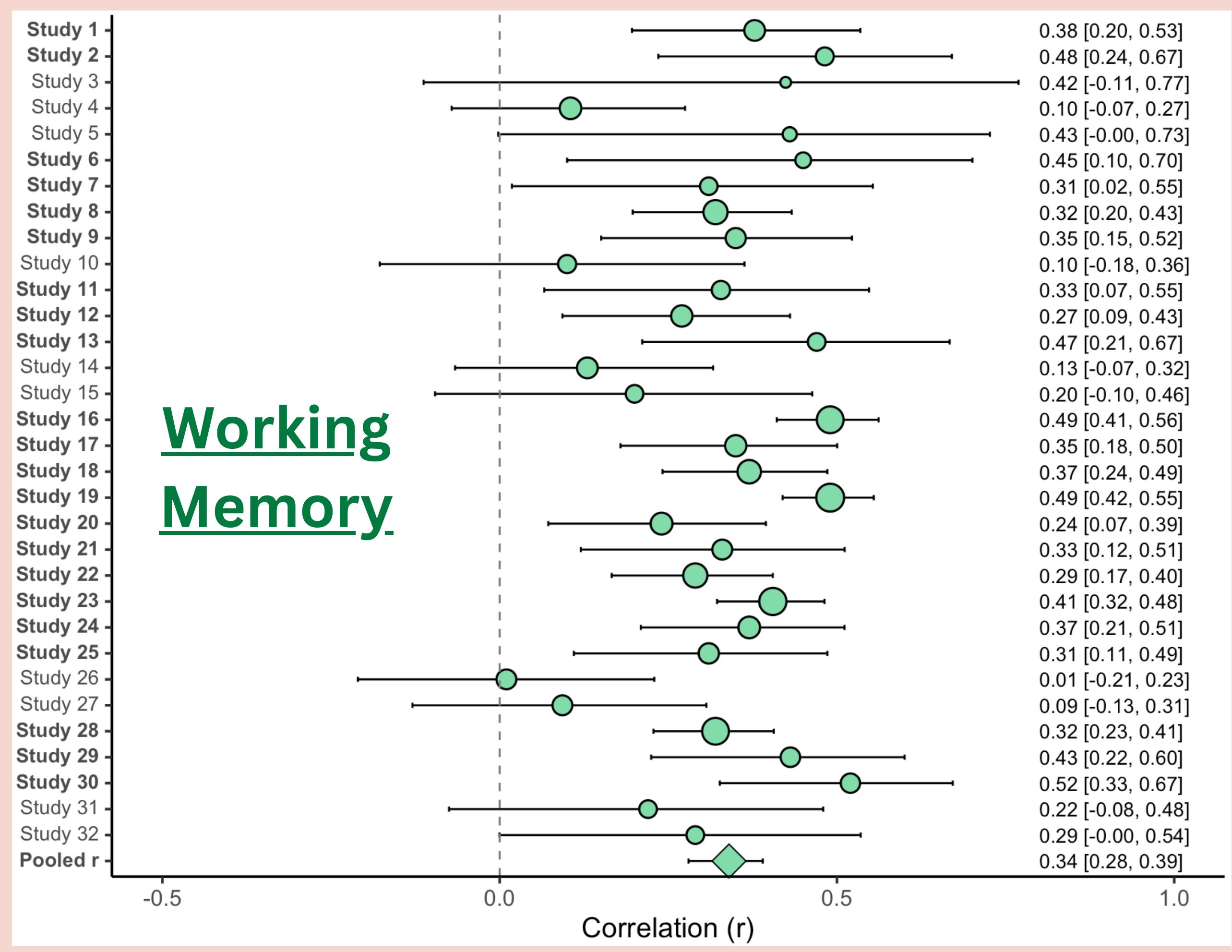
- ***Passage Comprehension:*** Multiple-Choice, Short Answer, Free Response
- ***Sentence Completion***
- ***Picture-Story Matching***

EFs:

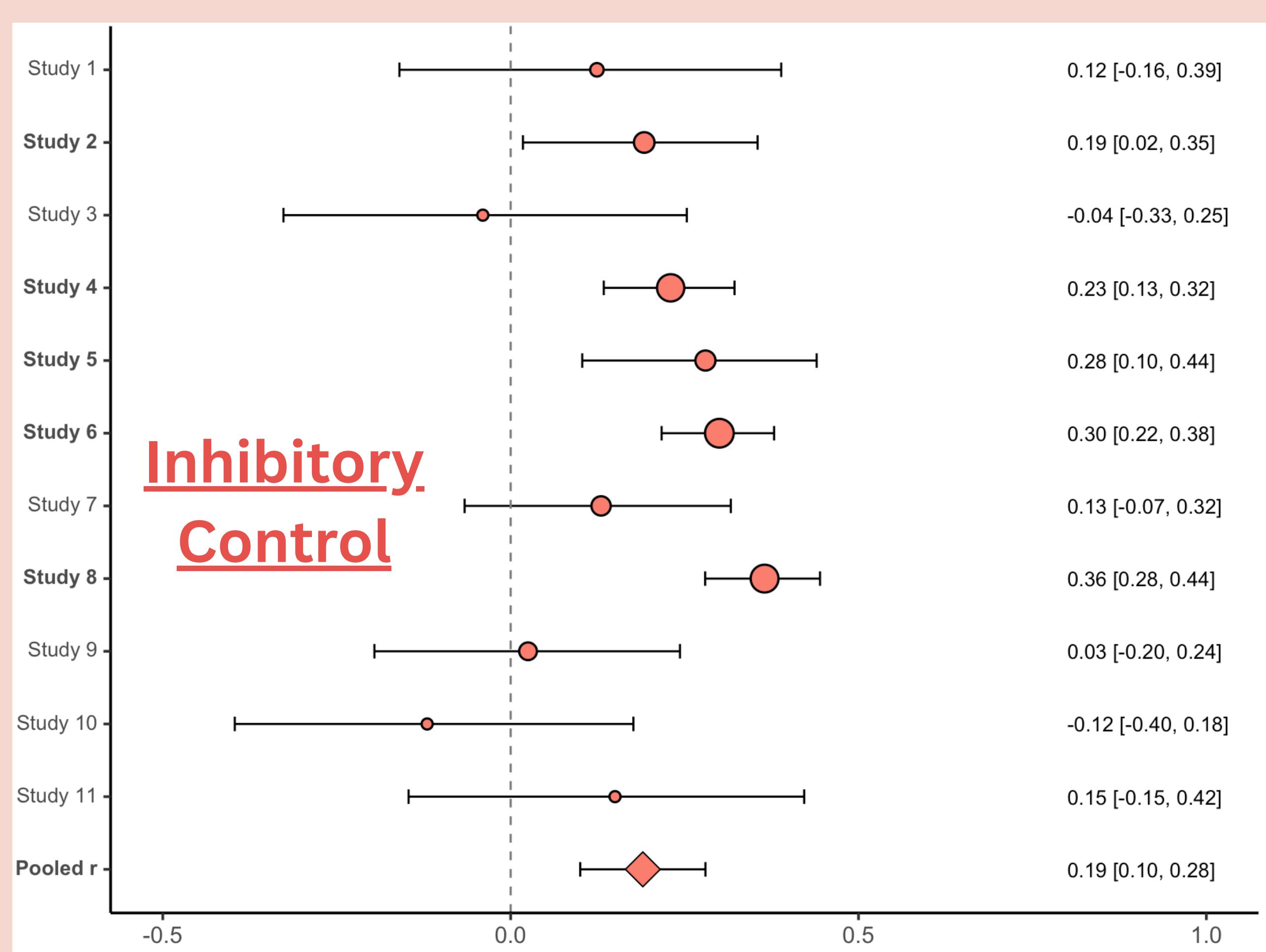
- ***Working Memory (WM)*:** Backward Digit Span, Complex Span
- ***Inhibitory Control (IC)*:** Simon Task, Flanker Task
- ***Cognitive Flexibility (CF)*:** Attention Shifting, Switching

Results

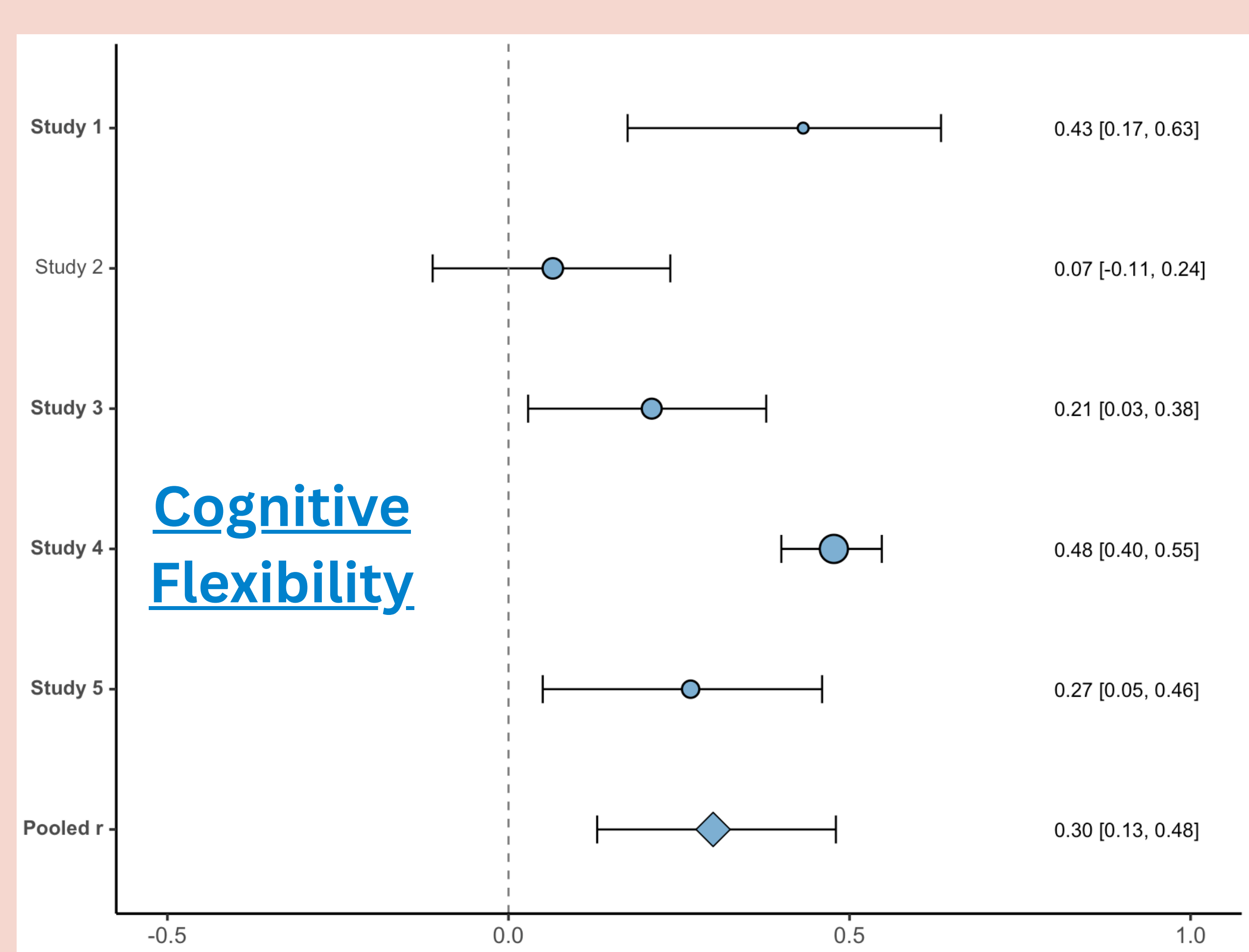
Correlations of each EF subcomponent with L2 RC. Bolded = significant; Dot size = sample size weight; Pooled *r* = random-effects meta-analysis estimate.



N = 4,172, Pooled *r* = .34, 95% CI [.28, .39], $p < .001$
I^2 = 57.23%, τ^2 = 0.01, Q(31) = 73.33, $p < .001$



N = 1,916, Pooled *r* = .19, 95% CI [.10, .28], $p < .001$
I^2 = 64.70%, τ^2 = 0.01, Q(10) = 25.73, p = .004



N = 797, Pooled *r* = .31, 95% CI [.13, .48], $p < .001$
I^2 = 79.00%, τ^2 = 0.03, Q(4) = 24.68, $p < .001$

Discussion

- Meta-analytic models confirmed significant positive associations with L2 RC across all three EFs
- WM was found to be the most frequently examined EF subcomponent in L2 RC, followed by IC, then CF
- Findings support claims that L2 RC is an active process that requires cognitive coordination, in addition to linguistic capacity

Future Directions

Due to the correlational nature of this work, we could not examine 1) how EFs and L2 RC reciprocally influence each other over time, and 2) the indirect relation between EFs and L2 RC. Future works could:

- Empirically explore how EFs and RC longitudinally support each other across the developmental trajectory
- Use meta-analytic structural equation modeling (MASEM) to simultaneously investigate relations among multiple linguistic and nonlinguistic constructs, to better decipher the direct and indirect roles of EFs on L2 RC

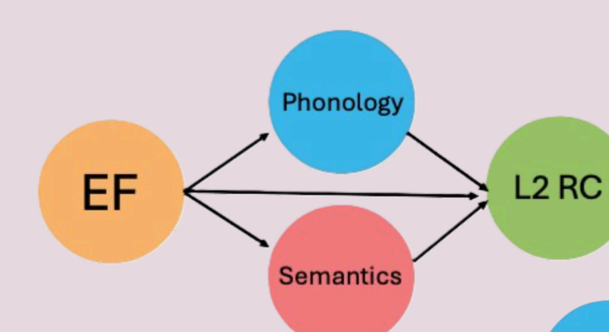


Figure 1.
Direct EF → RC model

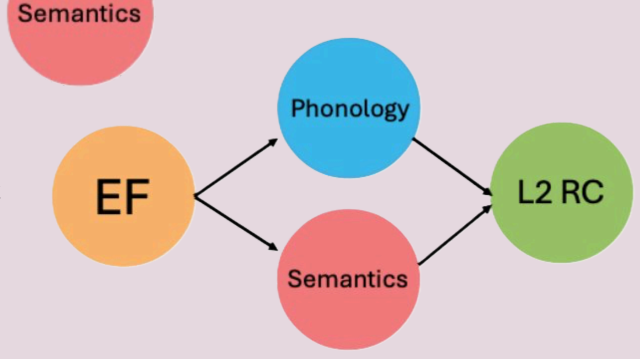


Figure 2.
Indirect EF → RC model

Conclusion

EFs should be treated more centrally in future reading models; much more work is warranted to better understand the relation between EFs, reading skill, and language profiles.

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